

Circle Theorems

GCSE Maths · Practice Questions

20 questions · 62 marks total

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Section A · Foundation

Questions 1–3 are the most straightforward; they get harder as you go on. Question 10 is the most challenging.

1 [1 mark]

What is the name of a straight line drawn from the centre of a circle to a point on the circumference?

.....

2 [2 marks]

- (a) What is a chord of a circle?
- (b) What is a tangent to a circle?

.....

3 [2 marks]

AB is a diameter of a circle. C is another point on the circle.
Write down the size of angle ACB and give a reason.

.....

4 [2 marks]

A, B and C are points on a circle with centre O. Angle AOC = 100° and B lies on the major arc.
Work out angle ABC.

.....

5 [3 marks]

PQRS is a cyclic quadrilateral. Angle PQR = 105° .
Work out angle PSR and give a reason.

.....

6 [3 marks]

A tangent touches a circle of radius 5 cm at point T. P is a point on the tangent with $PT = 12$ cm. O is the centre.

Work out the length of OP.

7 [3 marks]

Two angles at the circumference of a circle both stand on the same chord AB and are on the same side of it. One of them is 38° .

Write down the size of the other angle and give a reason.

8 [3 marks]

A and B are points on a circle with centre O. Angle $AOB = 70^\circ$.

Work out the size of angle OAB, explaining your reasoning.

9 [4 marks]

A, B and C are points on a circle with centre O, with C on the major arc AB.

Angle $OAB = 25^\circ$.

- (a) Write down angle OBA, with a reason.
 - (b) Work out angle AOB.
 - (c) Work out angle ACB.
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10 [4 marks]

AB is a diameter of a circle with $AB = 13$ cm. C is a point on the circle with $AC = 5$ cm.

- (a) Write down angle ACB, with a reason.
 - (b) Work out the length of BC.
 - (c) Work out the area of triangle ABC.
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Section B · Higher

Higher-tier style. Again, questions 1–3 are the gentlest and question 10 is the hardest.

1

[2 marks]

The angle at the centre of a circle standing on an arc is 84° .

Work out the angle at the circumference standing on the same arc.

2

[2 marks]

In a cyclic quadrilateral, two opposite angles are x° and $3x^\circ$.

Work out the value of x .

3

[3 marks]

PT is a tangent to a circle with centre O, touching at T. The radius is 6 cm and $PT = 8$ cm.

(a) Work out the length of OP.

(b) Work out the size of angle TPO, to 1 decimal place.

4

[3 marks]

A tangent touches a circle at A. A chord AB is drawn, and the angle between the tangent and the chord is 54° .

Write down the size of the angle in the alternate segment, and state the theorem you have used.

5

[4 marks]

ABCD is a cyclic quadrilateral in which AB is a diameter of the circle.

Angle $ADC = 130^\circ$.

(a) Work out angle ABC, with a reason.

(b) Write down angle ACB, with a reason.

(c) Work out angle BAC.

6

[4 marks]

A, B and C are points on a circle with centre O. Angle $BAC = 35^\circ$.

- (a) Work out angle BOC, with a reason.
 - (b) Work out angle OBC.
-

7

[4 marks]

TA is a tangent to a circle, touching at A. ABCD is a cyclic quadrilateral.

Angle $TAB = 65^\circ$ and angle $ABC = 80^\circ$.

- (a) Work out angle ACB, with a reason.
 - (b) Write down angle ADB, with a reason.
 - (c) Work out angle ADC.
-

8

[4 marks]

PT is a tangent to a circle with centre O, touching at T. The radius is 7 cm and $OP = 25$ cm.

- (a) Work out the length of PT.
 - (b) Work out angle TOP, to 1 decimal place.
-

9

[4 marks]

PA and PB are tangents to a circle with centre O, touching at A and B.

Angle $APB = 48^\circ$.

- (a) Write down angle OAP, with a reason.
 - (b) Work out angle AOB.
 - (c) Work out angle OAB.
-

10

[5 marks]

Prove that the angle at the centre of a circle is twice the angle at the circumference when both stand on the same arc.

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